

Group 4: CIKD++-RT: Knowledge-guided Visual Contradiction Evidence for Fine-grained Multimodal Misinformation Detection

Le Cao Xuan Mai, Phan Vu Minh Hang, Nguyen Minh Quan,
Le Huynh Minh Thi, Nguyen Do Gia Han, and Vu Nam Khanh

Abstract—We investigate the problem of fine-grained multimodal misinformation detection, in which evidence from text, images, and knowledge graphs (KG) must be analyzed jointly. Current KG-enhanced systems often utilize the KG merely as a passive context, whereas content-knowledge inconsistency requires an explicit contradiction signal. We propose CIKD++-RT, a residual TVCS-Transformer architecture that employs KG-guided visual patch evidence to refine the passive Text+Image+KG anchor. On the locked test split of FineFake kg_complete, the final variant no_c_emb improved Accuracy, Macro-F1, Weighted-F1, and CK-F1 metrics compared to the passive model, while achieving AUC = 0.726705 in the TVCS Real vs. CK task. The post-hoc temperature scaling method helps reduce Expected Calibration Error (ECE) without altering the model’s predictions. We position this result as a capability for knowledge-grounded inconsistency detection, rather than aiming for truth understanding or achieving SOTA across the entire task.

Index Terms—Fake News Detection, Knowledge Graph, Multimodal Inconsistency, Residual Transformer.

I. INTRODUCTION

A. Background

In the era of rapidly advancing technology, the explosion of multimodal fake news and misinformation on social media platforms has become a serious threat to the public and social stability. Modern fake news strategies increasingly exploit the complex combination of text and images to mislead readers. To systematically address this challenge, recent academic studies have shifted focus from traditional binary fake news classification to the problem of fine-grained misinformation detection [1], [2]. A foundational dataset in this domain is the FineFake benchmark, which standardizes the problem into an extremely rigorous 6-way classification task. This schema categorizes digital content into six distinct dimensions: real content (Class 0), text-image inconsistency (Class 1), content-knowledge inconsistency (Class 2), text-based fake news (Class 3), image-based fake news (Class 4), and other types (Class 5) [3]. Accurately identifying these fine-grained classification labels is a prerequisite for building targeted mitigation strategies corresponding to the core nature of each deceptive signal.

B. Importance of Knowledge Graph Evidence (KG Evidence)

Among the fine-grained classification categories, detecting Class 2 - Content-Knowledge Inconsistency (CK) poses the greatest technical challenge. Standard multimodal verification frameworks today typically optimize only for semantic alignment or similarity measurement between text and images.

Although this method is effective for identifying surface deviations in Class 1 (Text-image inconsistency), they are entirely powerless against sophisticated manipulation: when both text and image are consistent with each other but jointly violate factual knowledge. For example, a fabricated news piece might use a coherent description accompanied by an unedited original image, but effectively mislinks an entity to a false historical event or geographical location. Therefore, to robustly detect the CK inconsistency subset, the system must integrate an external structured knowledge source, specifically a Knowledge Graph (KG), to act as a ground-truth reference plane for cross-modal verification [4].

C. Limitations of Passive Knowledge Graph Fusion Paradigms

Previous knowledge-enhanced fake news detection models often integrated KGs in the form of static contextual features or conventional modality alignment. In typical architectures like KEAN [5] and KAlign [5], entity embeddings and relational triplets are processed passively and then fused with text/image representations via concatenation, bilinear pooling, or cross-attention. We define these methods as the passive KG fusion paradigm.

The core limitation of passive fusion is blending heterogeneous modalities into a common representation space but lacking an explicit supervisory objective to quantify the contradiction signal between visual content and the factual knowledge base. Because the neural network is not forced to directly score the conflict between visual patches and corresponding KG relations, the subtle indicators of factual deception are easily diluted during global fusion. Consequently, current systems achieve suboptimal performance when isolating content-knowledge inconsistency samples (Class 2: CK) from factual data.

D. Proposed Solution

To resolve this methodological bottleneck, we propose CIKD++-RT (the no_c_emb variant), a multimodal residual correction framework to explicitly verify and exploit the Text-Visual Contradiction Scoring (TVCS) signal. Instead of entirely replacing existing architectures, our solution acts as an additive correction layer on top of a passive neural network foundation.

Specifically, the system establishes a frozen baseline model to compute initial multimodal representations from text

(RoBERTa) [6], global image (CLIP), and entity knowledge (KG), acting as a passive Text+Image+KG anchor. On top of this anchor, a specialized TVCS model is integrated, utilizing KG relational queries ($W_q z_k$) to compute local cross-attention directly on CLIP visual patches (V_{patch}) [7]. This process renders a rich, multidimensional visual evidence feature (z_v) to capture explicit knowledge conflicts [8]:

$$z_v = \sum \text{softmax}(QK^T)V \quad (1)$$

The evidence feature z_v is then fed into the Residual Transformer architecture [8] to compute the correction logits (logit delta). The final inference process is formalized through a linear combination of the base logits and the residual correction part [9]:

$$\text{logits}_{\text{final}} = \text{logits}_{\text{base}} + \alpha \cdot \text{logits}_{\text{delta}} \quad (2)$$

Based on comprehensive diagnostic experiments, the final architecture completely eliminates the scalar confidence embedding vector (c_emb) from predecessor versions. Experiments indicate that directly injecting the scalar c_emb easily causes optimization turbulence, whereas preserving the multidimensional visual evidence feature z_v yields superior generalization capability on the target task.

E. Terminology and Definitions

To ensure clarity throughout this paper, we summarize key abbreviations and notations:

- **KG (Knowledge Graph) / CK (Content-Knowledge Inconsistency):** KG acts as external factual evidence. CK refers to the specific fake news category (Class 2) where content contradicts the KG.
- **TVCS (Text-Visual Contradiction Scoring):** Our proposed module that extracts conflict signals between KG queries and visual patches.
- **ECE (Expected Calibration Error):** A metric quantifying the gap between the model’s confidence and its true accuracy.
- **Macro-F1 / Weighted-F1 / CK-F1:** Evaluation metrics addressing class imbalance. Macro-F1 averages class F1s equally, Weighted-F1 accounts for class size, and CK-F1 is the specific F1 for the crucial CK class.
- **no_c_emb:** The final model variant that excludes scalar confidence embedding to reduce optimization noise.
- **z_v :** The rich multidimensional visual evidence feature output by TVCS.
- **logits_{base} / logits_{delta} / logits_{final}:** The raw predictions from the passive anchor model, the residual correction term, and the final combined output, respectively.

F. Contributions

The core technical and empirical contributions of this research include:

- **Explicit TVCS Mechanism:** Proposing the Text-Visual Contradiction Scoring (TVCS) mechanism, using structural knowledge relations to guide attention toward visual patches, directly extracting targeted contradiction evidence.

- **no_c_emb Residual Architecture:** Introducing an additive residual correction structure to refine the passive multimodal anchor. Experiments prove that preserving the rich multidimensional visual evidence feature z_v eliminates optimization noise from injecting the scalar confidence (c_emb) of previous systems.
- **Rigorous Locked Test Protocol:** Applying an evaluation procedure on a locked test split combined with a cache audit on the kg_complete data split of FineFake. The protocol applies secure benchmark design standards to ensure absolute zero data leakage and completely eliminates test-tuning phenomena.
- **Extensive Empirical Validation:** The final F4 model achieves 58.3140% Accuracy, 46.9755% Macro-F1, and a specialized CK-F1 score of 37.5546% for the knowledge inconsistency class on the locked test set. The use of Macro and CK-F1 metrics strictly adheres to evaluation principles on class imbalance distributions. The TVCS module achieves a Real-vs-CK separation AUC = 0.726705. Finally, the post-hoc temperature scaling method [10] optimizes confidence and system risk analysis, reducing Expected Calibration Error (ECE) from 14.80% to 3.85% without altering the argmax classification boundaries.

II. RELATED WORK

A. Multimodal Misinformation Detection

The rapid evolution of social media platforms has profoundly altered the landscape of deceptive content mining, shifting the research focus from traditional plain text analysis to multimodal misinformation detection [11]. Early methodologies in this subfield primarily relied on late fusion techniques or early concatenation directly between text and image feature representations. With the explosion of deep representation learning models, modern structural frameworks have shifted to unified neural backbones to simultaneously encode both structural and semantic properties across heterogeneous data streams [8], [12].

To systematically process these isolated inputs, standard architectures typically deploy joint attention-based systems to capture cross-modal correlations [13]. Although binary classification setups (e.g., merely separating general fake news from mainstream content) laid the foundational groundwork for this field, they inherently suffer from coarse-grained abstraction penalties.

Therefore, recent academic shifts emphasize the urgency of fine-grained misinformation modeling [14]. This classification schema is a prerequisite to isolate each individual deceptive dynamic, such as semantic manipulations confined strictly to text, deep-faked visual elements, or structural deviations from objective factual reference planes.

B. Knowledge-Aware Misinformation Detection

To overcome the limitations of perception-only architectures, integrating structured Knowledge Graphs (KGs) has become a focal research paradigm [15]–[17]. This method

maps multimodal tokens to unified entity networks (Wikidata, Wikipedia) to verify real-world factual claims [18]. Through the entity linking mechanism, the system extracts external relational implications and background knowledge triples to fortify the text and visual embedding spaces.

Pioneering studies often use graph embedding algorithms (TransE) or Graph Neural Networks (GNNs, GAT) to propagate relational signals across textual items and background entities [19]–[21]. Here, specialized attention channels are configured to compute dynamic weights for neighbor selections from the graph, ensuring optimal integration of relevant contexts and noise filtering.

However, traditional knowledge-enhanced models (such as KEAN, KGAlign, KAMP) all encounter a methodological bottleneck: passive KG fusion [5]. They blend external triplet properties from the KG into a generalized multimodal feature pool via global cross-attention or concatenation. This process dilutes factual deception indicators and the directional properties of structural knowledge, failing to isolate targeted friction components between visual patches and the semantic relations of the graph. Consequently, the system achieves suboptimal performance when isolating the content-knowledge inconsistency subset (Class 2: CK) under rigorous locked-test protocols.

C. Cross-Modal Inconsistency and Ambiguity Detection

Cross-modal inconsistency detection frameworks often optimize directly for identifying points of divergence, where language tokens contradict or directly clash with the factual scene depicted in associated image patches. To address this structural vulnerability, co-attention networks are widely deployed to simultaneously map image bounding regions and textual phrases into a common interaction space, helping to isolate local anomalies [14]. Advanced setups often apply contrastive learning objectives to widen the divergence threshold, assigning a high loss penalty to entities leaking misinformation [22], [23].

However, current cross-modal inconsistency detection systems rely on an implicit assumption: the reference reality is entirely self-contained within that local text-image pair. They solely focus on measuring surface compatibility between text and image [24]–[26]. Consequently, these models are completely powerless against sophisticated fabrications: the correlation between text and image achieves absolute surface consistency (Class 1 remains undetected), but their overall logical essence violates the objective knowledge of the external world (Class 2: CK). Lacking an external truth reference plane, merely optimizing internal similarity metrics significantly degrades the model’s generalization capability on complex real-world datasets.

D. Foundation Models and Base Feature Representation Learning

The bounded performance of complex multimodal classification systems is tightly constrained by the geometric structure of the neural networks acting as feature extraction backbones.

Modern deep learning optimization pipelines heavily rely on pre-trained large language models and vision-language foundation models to compute context-aware embeddings [27].

For the language feature space, improved variants of Masked Language Models like BERT or RoBERTa serve as the normative configuration due to their dynamic multi-layer bidirectional encoding characteristics [6], [7]. This mechanism allows the system to capture detached contexts from both the left and right sides of a token in a sentence, optimizing semantic representation for short and fragmented social media posts.

Humans communicate via text, but sophisticated fake news often manipulates image patches to deceive visually. Therefore, computer vision sub-modules have shifted aggressively from traditional Convolutional Neural Networks (CNNs) to Vision-Language Transformers [9], [28].

For instance, the CLIP architecture leverages large-scale contrastive learning objectives to globally align semantic scenes or extract high-resolution sequences of local visual patches (V_{patch}) [9], [28]. By processing images as sequences of operands akin to text, vision Transformers allow downstream models to directly compute cross-attention maps between structured knowledge entities and specific visual regions.

Additionally, inheriting additive residual connections remains a standard paradigm in deep network operations. The Residual Learning principle from the classic ResNet structure stabilizes gradient propagation, preventing the vanishing gradient phenomenon as neural networks progressively increase in depth [8].

In modern multimodal optimization tasks, this mechanism is extremely critical: it permits the integration of corrective pathways in the form of $\text{logits}_{\text{base}} + \alpha \cdot \text{logits}_{\text{delta}}$ to directly refine fixed, pre-frozen multimodal anchors, thereby preserving generalization features without disrupting the base representation space [29].

E. Reliability, Calibration, and Statistical Pitfalls in Model Evaluation

Evaluating deep classification models poses stringent challenges regarding empirical calibration and statistical pitfalls under skewed distributions. Deep neural networks often suffer from overconfidence in erroneous argmax predictions [10]. To quantify this risk, metrics such as Expected Calibration Error (ECE), Negative Log-Likelihood, and Brier Score are established to measure the divergence between predicted confidence and actual accuracy frequencies [10], [30], [31].

To lower the ECE metric without shifting argmax classification boundaries, the post-hoc temperature scaling calibration algorithm is deployed to reconstruct the global logit distribution via a single parameter $T > 0$ [10]:

$$\hat{p}_i = \max_k \sigma \left(\frac{z_i}{T} \right)_k \quad (3)$$

Alongside calibration, benchmark design demands rigorous protocols to avoid statistical pitfalls. When evaluating the minority knowledge inconsistency class (Class 2: CK) on imbalanced datasets, relying on the global Accuracy score

causes severe bias. Therefore, adhering to evaluation principles based on Macro-F1 and class-specific F1 scores (F1_{class}) is imperative [32].

Finally, to prevent data leakage and overtuning on the test set, the system applies a locked-test protocol paired with a strict cache audit procedure, guaranteeing the realistic generalization of the model.

F. Summary of Research Gaps

The core limitation of prevailing Knowledge Graph (KG) enhanced models is applying a passive KG fusion mechanism [5]. This method blends external knowledge properties into a global feature pool via global attention operations, diluting factual deception indicators and failing to isolate the explicit conflict signal between visual patches and the graph’s semantic relations. As a result, the system achieves suboptimal performance when confronting the content-knowledge inconsistency subset (Class 2: CK).

To address this gap, CIKD++-RT (the no_c_emb variant) acts as an additive residual correction layer upon a frozen network foundation [29]. The proposed framework employs the TVCS mechanism to guide KG relational queries to interact via local attention directly on individual visual patches, explicitly extracting the multidimensional inconsistency evidence feature z_v [33]–[37]. The model optimizes Real-vs-CK separation capability based on a rigorous locked-test protocol and post-hoc probability calibration, establishing a transparent validation basis for the realistic reliability of the downstream classification model.

III. DATASET, PROTOCOL, AND FEATURE CACHE

A. Dataset Description

We conduct experiments on FineFake [38], a public benchmark for fine-grained multimodal misinformation detection. FineFake is designed to move beyond binary fake/real classification by providing rich, structured annotations to distinguish various types of misinformation at a fine-grained level. Each sample in FineFake includes a text claim, an associated image, and a knowledge graph (KG) entity embedding retrieved from relevant entities in an external knowledge base.

1) *Fine-Grained Label Taxonomy*: FineFake applies a 6-way classification schema, where each sample is assigned one of the following labels:

This taxonomy captures a broader spectrum of misinformation compared to binary settings, specifically isolating **Content-Knowledge Inconsistency (CK, Label 2)** as a distinct category - the exact type of misinformation CIKD++-RT is explicitly designed to detect through knowledge-guided visual contradiction evaluation.

2) *Data Sources and Topics*: The complete FineFake dataset contains **16,909 samples** collected from seven online platforms: Snopes (44.69%), Reddit (23.94%), CNN (13.66%), The Washington Post (6.16%), Twitter (5.63%), AP News (4.34%), and CDC.gov (1.59%). The samples cover six topic domains: Politics (33.87%), Society (23.30%), Entertainment (21.88%), Conflict (10.16%), Business (5.93%), and Health

(4.20%), with a small fraction (0.67%) remaining uncategorized.

B. Dataset Statistics and Class Distribution

Table 2 summarizes the class distribution across the full dataset and its subsets (described in Section 3.3). Out of 16,909 total samples, **Real (Label 0)** is the largest class (44.37%), followed by **Text-based Fake (Label 3)** with 20.37%. **Content-Knowledge Inconsistency (CK, Label 2)** accounts for 9.53% of the total dataset (1,612 samples), while **Other (Label 5)** is the most underrepresented class with only 1.77% (300 samples). This significant **class imbalance** is a major challenge addressed through our validation-based model selection and calibration procedures (see Section 5).

C. Data Preprocessing Pipeline and Protocol Stratification

To ensure robust model training and avoid data leakage, we implement a strict sequential preprocessing pipeline: **Raw FineFake** → **data audit** → **KG missingness check** → **KG-complete protocol** → **TVCS-eligible protocol** → **feature cache** → **NaN/Inf/alignment checks**.

1) *Data Audit and KG Missingness*: A critical characteristic of the FineFake dataset is **the incomplete coverage of knowledge graph embeddings**. The raw FineFake dataset contains exactly 16,909 samples. Our initial data audit confirms that image existence is 100%, missing text is 0%, and missing image path is 0%. However, the audit reveals that **4,041 samples (23.8985%)** have missing or zeroed knowledge embeddings. The KG missingness rate is highly skewed across label categories: Label 4 (Image-based Fake) suffers the highest missing rate at 50.78%, while Label 5 (Other) has the lowest at 8.67%. For the CK class (Label 2), the missing rate is 24.69%.

Note: Because our proposed TVCS mechanism strictly requires complete knowledge graph embeddings and valid relations to function, we establish a final strict **kg_complete protocol** containing exactly **12,786 samples retained**. Consequently, **4,123 samples are excluded** from the final KG-grounded protocol (primarily due to the 4,041 missing KG embeddings, alongside a small number lacking valid relations or entity IDs). These excluded samples are not deleted from the original dataset; they are retained in the full manifest and cache for audit and future work. All analyses, final model training, and F4 evaluation are conducted strictly on the **kg_complete** subset.

The train/val/test splits were generated using stratified sampling (stratified by fine-grained labels), with a fixed random seed (seed=42) and a 70/10/20 ratio on the full dataset prior to KG filtering. In the kg_complete validation set, there are **120 CK samples**, and in the test set, there are **236 CK samples** - both exceeding the minimum threshold of 30 samples required for reliable CK class evaluation.

Critical Distinction: When referring to dataset statistics in this paper, we explicitly distinguish between the raw full dataset (N = 16,909) and the kg_complete subset (N = 12,786). All reported model performance metrics are evaluated exclusively on the kg_complete test split. Table IV details the exact post-filter class distribution for this subset.

Table I: Fine-grained label taxonomy of the FineFake dataset

Label ID	Category Name	Description
0	Real	Content is consistent and shows no signs of misinformation.
1	Text-Image Inconsistency	There is a mismatch or contradiction between the text claim and the accompanying image.
2	Content-Knowledge Inconsistency (CK)	Content contradicts verifiable information from an external knowledge graph - the primary target category of our system.
3	Text-based Fake	Misinformation is primarily carried by the text claim.
4	Image-based Fake	Misinformation is primarily fabricated or manipulated in the image.
5	Other	Misinformation cases that do not clearly fit into the above categories.

Table II: Class distribution across the raw full FineFake dataset (N = 16,909)

Label ID	Category	Total Count	% of Total
0	Real	7,502	44.37%
1	Text-Image Inconsistency	1,477	8.74%
2	Content-Knowledge Inconsistency (CK)	1,612	9.53%
3	Text-based Fake	3,444	20.37%
4	Image-based Fake	2,574	15.22%
5	Other	300	1.77%
—	Total	16,909	100%

Table III: Statistics of the kg_complete subset

Split	Count	% of kg_complete
Train	8,900	69.61%
Validation	1,300	10.17%
Test	2,586	20.22%
Total (kg_complete)	12,786	—

2) *TVCS Eligible Subset*: Another subset, **tvcs_eligible**, is used exclusively for training the TVCS expert module (Section 4). This subset includes all kg_complete samples with fine-grained labels of either **Real (Label 0)** or **Content-Knowledge Inconsistency (Label 2)** - the two classes defining the binary discrimination task for TVCS. The TVCS expert is trained to generate a contradiction score differentiating Real samples from CK samples.

Note that the tvcs_eligible subset is a strict subset of kg_complete, and the TVCS expert is never exposed to the locked test split during training (validation-only model selection protocol, detailed in Section 5).

D. Feature Cache

To enable efficient training across multiple model configurations and ablations without repeated inference through large pre-trained encoders, we first extract and cache all input features as NumPy arrays. The feature extraction pipeline processes the full dataset once and generates three cache directories: **full**, **kg_complete**, and **tvcs_eligible**, each containing aligned feature arrays for all samples in that subset.

The following features are extracted and cached for each sample:

All cached arrays underwent rigorous quality checks. No NaN or Inf values were detected in any feature arrays across all caches. The patch features were verified to have non-zero variance and differ from the global image features, confirming that spatial information is preserved. The relation vocabulary contains 1,029 unique relation types, with token 0 reserved for <pad>(samples without relations) and token 1 for <unk>.

The full feature cache for the kg_complete subset corresponding to N = 12,786 samples has the following shapes:

All feature arrays are aligned via a common **sample_ids** index and a **split_ids** array, ensuring that train (split_id=0), val (split_id=1), and test (split_id=2) splits can be accurately reconstructed from any cache without ambiguity.

E. Evaluation Protocol Safeguards

The following protocol safeguards are enforced in all experiments to ensure the integrity of our evaluation:

- 1) **Locked test split.** The test split is used exclusively for final evaluation reported in this paper (Section 6). During any phase of model development, architecture search, or hyperparameter tuning, the test split is unaccessed. This is strictly enforced in code via the split_id filter: the final evaluation phase (F4 Stage) loads test samples only after model training and selection are complete.
- 2) **Validation-only model selection.** All model checkpoints are selected based on their performance on the validation split, using a selection score defined as a weighted combination of validation Macro-F1 and CK-F1. The final model used for test set evaluation is exclusively the checkpoint that achieved the best validation selection score.
- 3) **No duplicate samples.** The dataset contains no duplicate sample IDs or duplicate image paths across splits. Each sample appears in exactly one of the train, validation, or test splits.
- 4) **No test tuning guarantee.** No hyperparameters - including learning rates, dropout, regularization weights, or loss function parameters - were tuned using feedback from the test set. All hyperparameter decisions were made on the validation split.
- 5) **Sanity check.** A KG+Relation sanity check model trained solely on KG and relation features achieved a validation Macro-F1 of **0.1713**, serving as a diagnostic sanity check confirming that the features alone (without

Table IV: Class distribution after KG filtering (kg_complete subset, N = 12,786)

Label ID	Category	Count	% of kg_complete
0	Real	6,298	49.26%
1	Text-Image Inconsistency	924	7.23%
2	Content-Knowledge Inconsistency (CK)	1,211	9.47%
3	Text-based Fake	2,848	22.27%
4	Image-based Fake	1,240	9.70%
5	Other	265	2.07%
—	Total	12,786	100%

Table V: Statistics of the tvcs_eligible subset

Split	Count	CK Samples	Real Samples
Train	5,241	855	4,386
Validation	762	120	642
Test	1,506	236	1,270
Total	7,509	1,211	6,298

architectural contributions) do not trivially solve the task.

IV. METHOD: CIKD++-RT

In this section, we present the architecture of the proposed CIKD++-RT (Contextual Information Knowledge Distillation with Residual Transformer) framework. We formulate the problem as a 6-class multimodal misinformation detection task, where the input $x = (T, I, K, R)$ consists of Text (T), Image (I), Knowledge Graph entities (K), and Relations (R), and the output $y \in \{0, 1, 2, 3, 4, 5\}$. A key focus in our research is the accurate identification of Content-Knowledge (CK) inconsistency, corresponding to class 2 in the problem formulation.

Our final model, the no_c_emb variant, relies entirely on the cross-attention mechanism to explicitly extract visual contradiction evidence, avoiding reliance on unreliable prior scalar quantities.

The overall data flow across these components can be summarized as follows:

- **Text Input:** Encoded into a text feature using RoBERTa.
- **Image Input:** Encoded into both a global visual feature and fine-grained spatial patch tokens via CLIP.
- **Knowledge Input:** KG entities and relations are encoded into structured representations.
- **TVCS Evidence Extraction:** The TVCS module actively uses the KG/relation representations as a query, attending to CLIP visual patches (key/value) to isolate contradiction signals, producing the rich visual evidence feature z_v .
- **Prediction Pipeline:** First, the passive anchor linearly combines text, global image, and KG to produce logits_base. Then, the Residual Transformer processes z_v (and other modalities) to compute the correction term logits_delta. The final prediction is formed via residual addition: $\text{logits_final} = \text{logits_base} + \alpha * \text{logits_delta}$. Notably, the final no_c_emb model fully removes the scalar class prior (c_emb) while explicitly preserving the TVCS multidimensional visual evidence z_v .

The system framework utilizes a passive Text+Image+KG anchor to generate logits_base. Simultaneously, the TVCS

expert module explicitly extracts contradiction evidence (z_v) via a cross-attention mechanism.

A. Passive T+I+KG Anchor

To establish a solid representation foundation, we first set up a passive fusion baseline. The text utterance T is encoded using RoBERTa, while the image I is processed via CLIP to extract both global and patch-level features. The KG entities K and relations R are encoded into continuous vector representations.

In the passive anchor configuration, these modalities are simply concatenated to create a combined multimodal representation. This combined representation is then passed through a fully connected layer to compute the base classification logits (logits_base). Although this approach integrates knowledge as supplementary context, it lacks the explicit mechanisms required to actively detect intentional contradictions between visual content and the provided knowledge.

B. TVCS: Knowledge-Guided Visual Contradiction Scoring Expert Module

To overcome the limitations of the passive fusion method, we introduce the TVCS (Knowledge-guided Visual Contradiction Scoring) expert module. The objective of this module is to model explicit interactions between external knowledge and visual components, thereby enabling the detection of multimodal inconsistencies at a fine-grained level.

The TVCS expert module uses a cross-attention mechanism, where the query is derived from KG and relation embeddings (z_k), while keys and values are extracted from CLIP visual patches (V_{patch}). This architecture allows the model to actively scan the image to find specific patches that support or contradict the provided knowledge claim. The attention operation yields a weighted visual evidence vector, z_v :

$$\text{query} = W_q z_k \quad (4)$$

$$\text{keys} = W_k V_{patch} \quad (5)$$

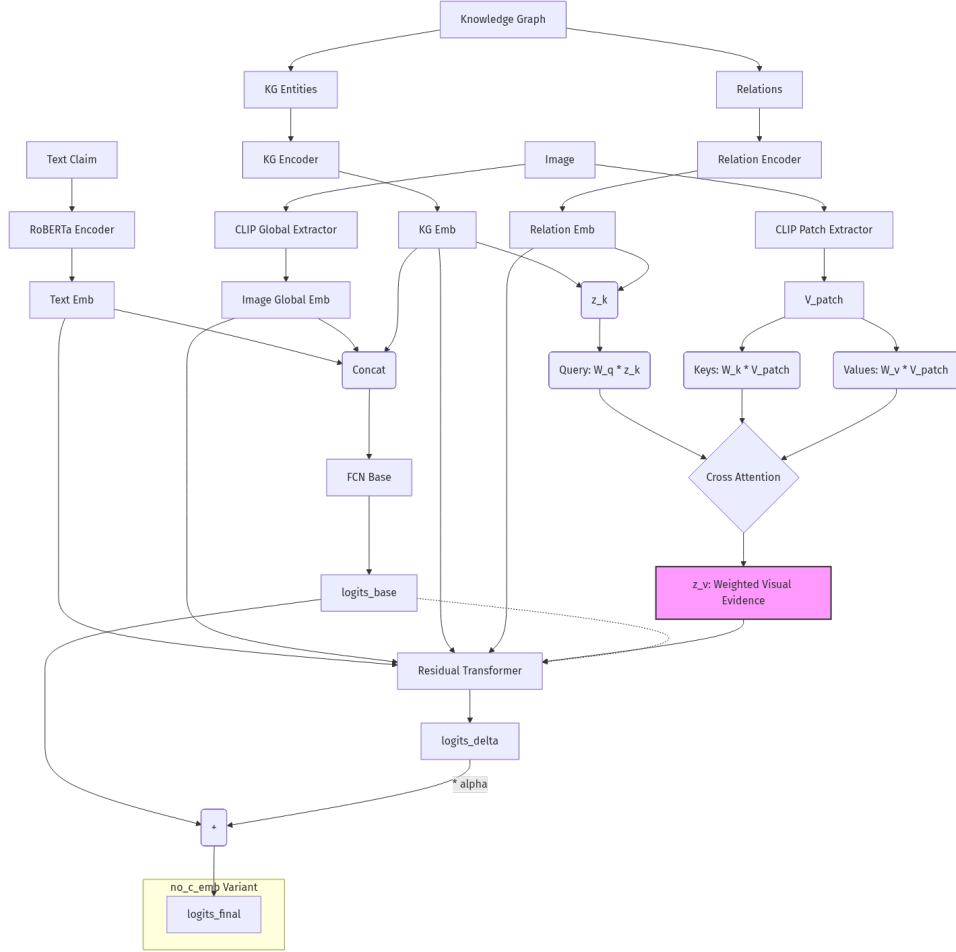
$$\text{values} = W_v V_{patch} \quad (6)$$

$$z_v = \sum \text{softmax}(\text{query} \cdot \text{keys}^T) \text{values}_i \quad (7)$$

It is important to note that the evidence z_v derived from TVCS captures visual signals correlated with knowledge inconsistency, acting as a knowledge-grounded inconsistency signal, rather than absolute proof of truthfulness.

Table VI: Extracted and cached features (kg_complete subset, N = 12,786)

Feature	Backbone Model	Shape	Description
text_features	RoBERTa-base [6]	[N, 768]	CLS token embedding representing the entire text claim
image_features_global	CLIP ViT-B/32 [39]	[N, 512]	Global image embedding after CLIP image projection
image_features_patch	CLIP ViT-B/32	[N, 49, 512]	Image embedding per patch (49 patches, CLS excluded)
kg_features	FineFake KG Entity Embeddings	[N, 100]	Knowledge graph embedding with 100 dimensions
relation_ids	FineFake Relation Field	[N]	Integer ID mapping the first relation type to a vocabulary of 1,029 relation types

**Figure 1:** Overall architecture of CIKD++-RT no_c_emb.**Table VII:** Feature cache shapes for the kg_complete subset (N = 12,786)

Feature	Shape (kg_complete)
text_features	[12786, 768]
image_features_global	[12786, 512]
image_features_patch	[12786, 49, 512]
kg_features	[12786, 100]
relation_ids	[12786]

C. Residual Transformer Correction

Instead of replacing the highly stable base classifier, we use the TVCS expert module as a residual correction branch.

The derived visual evidence z_v , along with the original text features, global image, and KG relations, are processed through a Residual Transformer to compute the correction vector logits_delta .

The final prediction is then formulated as a residual addition:

$$\text{logits_final} = \text{logits_base} + \alpha * \text{logits_delta} \quad (8)$$

This design ensures that the model preserves the generalized discriminative capability of the passive anchor, while dynamically adjusting predictions based on explicit contradiction evidence when necessary.

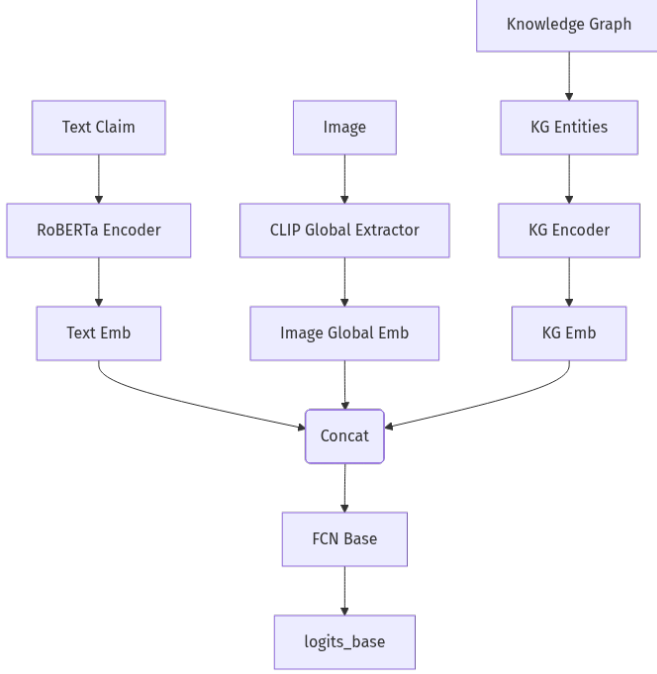


Figure 2: Passive Text+Image+KG anchor for computing logits_base.

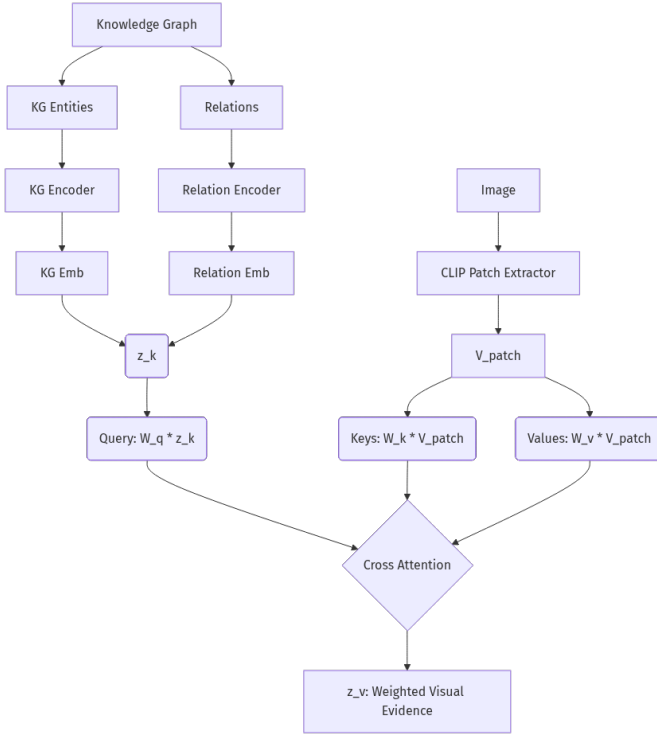


Figure 3: TVCS module for KG-guided visual evidence extraction.

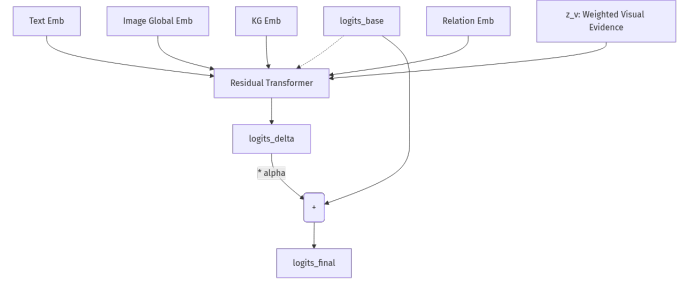


Figure 4: Residual Transformer correction and final logit composition.

D. no_c_emb Variant

During the architectural design phase, we initially experimented with integrating a scalar class embedding representation (c_emb) to directly inject class structure priors. However, empirical analysis (detailed in Section 6) revealed that c_emb acted as a detrimental "shortcut", encouraging the neural network to overfit based on prior class probabilities rather than genuinely contrasting multimodal evidence.

Consequently, our final proposed architecture is the no_c_emb variant. By completely eliminating the scalar quantity c_emb , we force the model to rely entirely on the high-quality visual evidence z_v derived from TVCS. The TVCS module remains the core, indispensable component in this final architecture.

E. Training Objective and Model Selection

The model is optimized using a combined loss function, comprising the standard cross-entropy classification loss (L_{cls}) and a specialized binary cross-entropy loss for TVCS (L_{tvcs}). The TVCS loss is strictly masked and applied exclusively to distinguish between genuine multimodal alignment (Real) and fabricated contradiction samples (CK).

$$L_{total} = L_{cls} + \lambda * L_{tvcs} \quad (9)$$

To prevent data leakage and ensure reliable generalization capability, the model checkpoint selection is strictly performed following a validation-only protocol, without any tuning on the test set. All final reported test metrics are obtained from a locked evaluation phase.

V. EXPERIMENTAL SETUP, EVALUATION METRICS, AND VALIDATION PROTOCOL

A. Class Imbalance Handling Strategy

Because FineFake is naturally imbalanced, properly handling class distribution is critical to system reliability. We deliberately **did not use** data-level balancing techniques such as oversampling, undersampling, SMOTE, or fake sample generation. Enforcing forced equal class balancing distorts the natural distribution of real-world misinformation and risks overfitting on very small classes (such as Label 5: "Other", which accounts for only 1.77% of the data).

Instead, our handling strategy strictly follows a protocol-first approach:

- 1) **Dataset/Protocol-level handling:** We strictly utilized stratified splitting based on fine-grained labels to ensure the minority classes (especially CK and Other) are proportionally represented across Train, Validation, and Test splits. We manually checked the CK ratio to ensure sufficient sample sizes in the evaluation sets.
- 2) **Evaluation-level handling:** Rather than relying solely on overall Accuracy (which heavily favors dominant classes), we explicitly evaluate using Macro-F1, Weighted-F1, per-class F1 (e.g., CK-F1), and confusion matrices to transparently observe the model’s true performance on minority categories.
- 3) **Model-level handling:** Model-level interventions (such as focal loss weighting or validation-based selection criteria targeting CK-F1) are applied only *after* the rigorous stratified data protocol is fixed, ensuring no leakage or synthetic distribution bias.

B. Baselines and Diagnostic Models

To comprehensively evaluate the effectiveness of the proposed CIKD++-RT architecture, we compare against a final set of baselines and diverse diagnostic models. Unimodal baselines include a **Text-only** model (RoBERTa) and an **Image-only** model (CLIP), acting as basic sanity checks. The fundamental multimodal baseline is **T+I concat**, which fuses text and image without external knowledge.

For knowledge-enhanced baselines, we evaluate **T+I+KG concat**, acting as the primary passive KG anchor. We also compare against stronger fusion architectures: **G0-E Transformer** (a highly stable Transformer-based passive fusion architecture) and **G1-B Co-attention** (a generalized multimodal cross-attention baseline). Finally, we include the **Old CIKD** (CKBoost-MoE) framework to demonstrate the evolutionary improvements of the current version.

Additionally, several models are strictly utilized for internal diagnostic and feasibility check purposes, including **CAFE-lite**, **Stage I**, and previous ablation variants. These models are strictly treated as diagnostic references, not final comparative baselines.

C. Evaluation Metrics

We evaluate all models using four complementary classification metrics, suitable for the imbalanced 6-way classification context of FineFake.

1) **Accuracy:** Accuracy measures the proportion of correctly classified samples over the total test set:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i] \quad (10)$$

where y_i is the ground truth label and $\hat{y}_i$ is the predicted label for sample i . Although Accuracy provides a simple overall performance measure, it can be misleading under severe class imbalance conditions. A model that ignores minority classes

(e.g., Label 5: Other, accounting for only 1.77% of samples) can still achieve high accuracy by performing well on the dominant class. Therefore, we report Accuracy alongside the additional metrics below.

2) **Macro-F1:** Macro-F1 computes the F1 score for each class independently, then takes the unweighted average, regardless of class frequency:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=0}^{C-1} F1^c, \quad \text{where} \quad F1^c = \frac{2 \cdot P^c \cdot R^c}{P^c + R^c} \quad (11)$$

with $C = 6$ classes, P^c and R^c denoting per-class Precision and Recall.

In imbalanced datasets like FineFake kg_complete (where Label 0 constitutes 49.2% of the training set and Label 5 only 2.1%), Macro-F1 penalizes models that exclusively perform well on majority classes. By treating all six categories equally, this metric ensures that improvements on challenging minority classes (especially Label 2: CK and Label 5: Other) are recognized. This makes Macro-F1 the most informative aggregate metric to assess whether CIKD++-RT genuinely enhances fine-grained discrimination or merely improves majority class accuracy [40].

3) **Weighted-F1:** Weighted-F1 calculates the F1 score for each class but weights each class’s contribution by the number of ground truth samples in the test set:

$$\text{Weighted-F1} = \frac{\sum_{c=0}^{C-1} n^c \cdot F1^c}{\sum_{c=0}^{C-1} n^c} \quad (12)$$

where n^c is the number of ground truth samples for class c in the test set. Weighted-F1 reflects overall performance weighted by the actual label distribution, and is reported alongside Macro-F1 to provide a complementary perspective: if Weighted-F1 is significantly higher than Macro-F1, it indicates the model primarily benefits from majority class performance rather than uniform improvement across all classes.

4) **CK-F1 (Label 2 F1 Score):** CK-F1 is the F1 score computed specifically for Label 2 (Content-Knowledge Inconsistency):

$$\text{CK-F1} = F1^{c=2} = \frac{2 \cdot P_{\text{CK}} \cdot R_{\text{CK}}}{P_{\text{CK}} + R_{\text{CK}}} \quad (13)$$

CK is the target class of the knowledge-guided contradiction detection mechanism in CIKD++-RT. Improving CK-F1 is the primary technical objective of the system. We report CK-F1 separately to make the model’s contribution to this specific detection task transparent and verifiable.

In the locked test evaluation, the CK-F1 improvement over the passive T+I+KG baseline is primarily driven by an **improvement in precision** (31.93% $\rightarrow$ 38.74%) and a **reduction in false positives** (194 $\rightarrow$ 136), while CK recall slightly decreases (38.56% $\rightarrow$ 36.44%). We report this trade-off honestly, reflecting the model’s mechanism of reducing false CK predictions rather than increasing coverage.

5) *TVCS AUC*: To evaluate the explicit contradiction scoring sub-problem, we measure the **TVCS AUC** for the capability to separate Real and CK samples. This metric assesses the TVCS expert module’s ability to generate continuous contradiction scores that distinguish genuine alignment (Real, Label 0) and content-knowledge inconsistency (CK, Label 2), independent of the final classifier.

D. Calibration Metric: Expected Calibration Error (ECE)

Beyond classification metrics, we evaluate the quality of the model’s confidence calibration using the Expected Calibration Error (ECE) [10]:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (14)$$

where samples are grouped into $M = 10$ equally spaced confidence bins B_m , and $\text{acc}(B_m)$, $\text{conf}(B_m)$ are the average accuracy and average predicted confidence in each bin, respectively.

A well-calibrated model will have an ECE near 0%, meaning softmax confidence scores faithfully reflect empirical accuracy. ECE is particularly critical in our context because the 6-way class imbalance can cause the model to generate overconfident predictions for the dominant class. We report ECE before and after calibration using the temperature scaling method described in Section 5.4.

E. Calibration Protocol: Temperature Scaling

We apply Temperature Scaling [10] as a post-hoc calibration method. Temperature scaling introduces a single scalar parameter $T > 0$ to rescale the logits before conversion to probabilities via softmax:

$$\hat{p}^c = \frac{\exp(z^c/T)}{\sum_{c'=0}^5 \exp(z^{c'}/T)} \quad (15)$$

where z^c is the raw logit for class c , and T is optimized by minimizing the Negative Log-Likelihood (NLL) on the **validation set**.

Key properties of temperature scaling:

- When $T > 1$: the distribution is "softened" - confidence decreases, predictions become less certain. This is suitable when the model is overconfident.
- When $T < 1$: predictions are sharpened.
- Temperature scaling **does not alter the argmax** of the output distribution. The predicted class label remains identical before and after calibration; only the confidence score changes.

Table VIII: Calibration results (H1-Cal canonical)

Dataset	Raw ECE	Calibrated ECE	Temperature T
Validation	14.876%	4.501%	1.522523
Test	14.80%	3.85%	1.522523

The optimal temperature $T = 1.522523$ is determined by minimizing NLL on the validation set. Applying this fixed T to the locked test set reduces ECE from **14.80%** to **3.85%**

with **no argmax changes** - confirming that all classification performance metrics (Accuracy, F1) are entirely unaffected by calibration. All calibrated confidence scores reported in this paper utilize this canonical H1-Cal temperature.

Throughout the paper, all classification metrics (Accuracy, Macro-F1, Weighted-F1, CK-F1) are calculated from raw argmax predictions and are unaffected by temperature scaling. Calibration exclusively impacts the reported confidence/probability values and ECE. The distinction between Raw ECE (14.80%) and Calibrated ECE (3.85%) is consistently maintained throughout.

F. Validation-Only Model Selection Protocol

All model checkpoints are selected based on a selection score computed **solely on the validation set**:

$$\text{Selection Score} = 0.5 \times \text{Macro-F1}_{\text{val}} + 0.5 \times \text{CK-F1}_{\text{val}} \quad (16)$$

This composite score equally balances overall multi-class performance (Macro-F1) and target CK class performance (CK-F1), reflecting the dual objective of the system. The checkpoint achieving the highest selection score on the validation set is used as the final model for test set evaluation.

Rationale for validation-based selection: Using test set performance to guide model selection would constitute test data leakage and invalidate the reported results. The validation selection protocol ensures the test set operates exclusively as a held-out split, providing an unbiased estimate of generalization performance.

G. No Test Tuning Guarantee Protocol

We enforce a strict no-test-tuning guarantee across all experiments:

- 1) **Hyperparameter Tuning:** All hyperparameters - including learning rate (1×10^{-4}), weight decay (1×10^{-4}), dropout rate (0.2), TVCS loss weight ($\lambda_{TVCS} = 0.5$), residual normalization ($\mu = 0.01$), residual limit ($\alpha_{max} = 0.5$), Transformer depth ($L = 2$ layers, $H = 4$ heads, $d_{\text{model}} = 256$), and focal loss $\gamma = 1.0$ - are fixed based on validation performance. No hyperparameters were adjusted following any observation of test set results.
- 2) **Early Stopping:** Training utilizes patience-based early stopping with patience = 5 epochs, triggered by the validation selection score. The test split is never loaded or accessed during training.
- 3) **Temperature Calibration:** The temperature scalar $T = 1.522523$ is optimized on the validation set NLL. This value is subsequently applied to the test set without any further adjustments.
- 4) **Locked Test Set Access:** In the codebase, the test split (`split_id = 2`) is loaded and evaluated exclusively in the F4 Stage (`stage_f4_final_lock.py`), which is executed once and guarded by an explicit `-execute_final` flag. All other stages operate exclusively on the train (`split_id = 0`) and validation (`split_id = 1`) splits.

- 5) **Seed Reproducibility:** All experiments employ seed = 42 for Python, NumPy, and PyTorch random number generators, ensuring complete reproducibility of the results.

H. Bootstrap Confidence Intervals

To assess the statistical reliability of metric improvements over the baseline, we report **paired bootstrap confidence intervals** [41] with 1,000 resamples (seed = 42). For each metric m and each bootstrap resample b :

$$\Delta_b(m) = m(\text{CIKD++-RT, resample}_b) - m(\text{Baseline, resample}_b) \quad (17)$$

The 95% confidence interval is calculated as the 2.5th and 97.5th percentiles of $\{\Delta_b(m)\}_{b=1}^{1000}$. We also report the probability of improvement $P(\Delta > 0)$ as the proportion of resamples where the model outperforms the baseline.

Interpretation of results:

- The improvement in Accuracy is well-supported statistically (98.6% probability).
- The Macro-F1 improvement is modest and positive (63.0% probability), but the 95% confidence interval crosses zero, indicating the improvement is not statistically conclusive.
- CK-F1 demonstrates a meaningful directional improvement (88.0% probability) aligning with the targeted TVCS mechanism, but the confidence interval also includes zero.

We do not claim these improvements as statistically conclusive. The bootstrap results are reported to transparentize the magnitude and uncertainty of the observed improvements.

VI. RESULTS

A. Main Results on Locked Test Split

We evaluate the final CIKD++-RT no_c_emb model (F4) against the passive T+I+KG anchor and other multimodal baselines on the locked test split. The F4 model achieves an Accuracy of 58.3140%, Macro-F1 of 46.9755%, and Weighted-F1 of 59.5127%. These results demonstrate an improvement over the passive fusion baseline across all key metrics. Most importantly, the system framework improves the targeted CK-F1 metric to 37.5546%.

To confirm that the problem cannot be solved trivially by feature engineering, a sanity check model trained solely on KG and relation features achieved a validation Macro-F1 of only 0.1713. This confirms that the architectural contributions of the TVCS-guided residual formulation are intrinsically necessary for the observed performance improvements.

B. TVCS Explicit Evidence Separation

The primary contribution of our architecture is the explicit derivation of visual contradiction evidence. Analysis of the continuous TVCS evidence scores reveals a clear and measurable separation between genuine alignment (Real) and fabricated multimodal contradictions (CK).

On the locked test split, the scalar TVCS achieves an AUC of 0.726705 for the Real-vs-CK separation task. The mean TVCS score for Real samples is 0.304671, whereas the mean TVCS score for CK samples is significantly higher at 0.503780 (a delta of 0.199109). This confirms that the cross-attention module successfully learned a knowledge-grounded inconsistency signal that directly correlates with actual discrepancies between text, image, and knowledge.

C. CK Classification Mechanism

While the overall CK-F1 metric improved, a detailed investigation into the mechanism driving this improvement is essential. The enhancements are entirely driven by a significant increase in classification precision. Specifically, CK precision increased from 31.93% (in the passive anchor) to 38.74% (in the F4 model). Correspondingly, the number of CK false-positive predictions decreased from 194 to 136.

We observe a slight decline in CK recall, dropping from 38.56% to 36.44%. This indicates that explicit visual evidence prompts the model to become more cautious and precise when flagging contradictions, effectively reducing false alarms with a minor trade-off in coverage.

D. Evidence from Ablation and Forensic Analysis

To isolate the contribution of the visual evidence, we conduct an ablation experiment where the TVCS evidence is zeroed out ($z_v = 0$). Under this condition, the model's Accuracy drops to 0.5619, Macro-F1 falls to 0.4357, and CK-F1 plummets sharply to 0.3080. Furthermore, disrupting the alignment by randomly shuffling the image patches collapses the Real-vs-CK AUC from 0.726705 to near-random levels (0.5448). These forensic diagnostics unequivocally validate that the residual improvements are directly derived from structurally grounded visual contradiction evidence.

E. Calibration Confidence

The raw F4 model exhibits a tendency towards extreme overconfidence, resulting in an initial raw Expected Calibration Error (ECE) of 14.80%. After applying the H1-Cal post-hoc temperature scaling procedure - where the optimal temperature $T = 1.522523$ is estimated via Negative Log-Likelihood on the validation set - the test set ECE significantly improves to 3.85%. Because this transformation monotonically scales the logits, this substantial enhancement in confidence reliability is achieved entirely without any changes to the argmax predictions.

F. Qualitative Analysis and Case Studies

To gain deeper insights into the internal workings of the TVCS module and its effect on the final predictions, we conduct a qualitative analysis on several representative samples from the locked test set of the FineFake dataset:

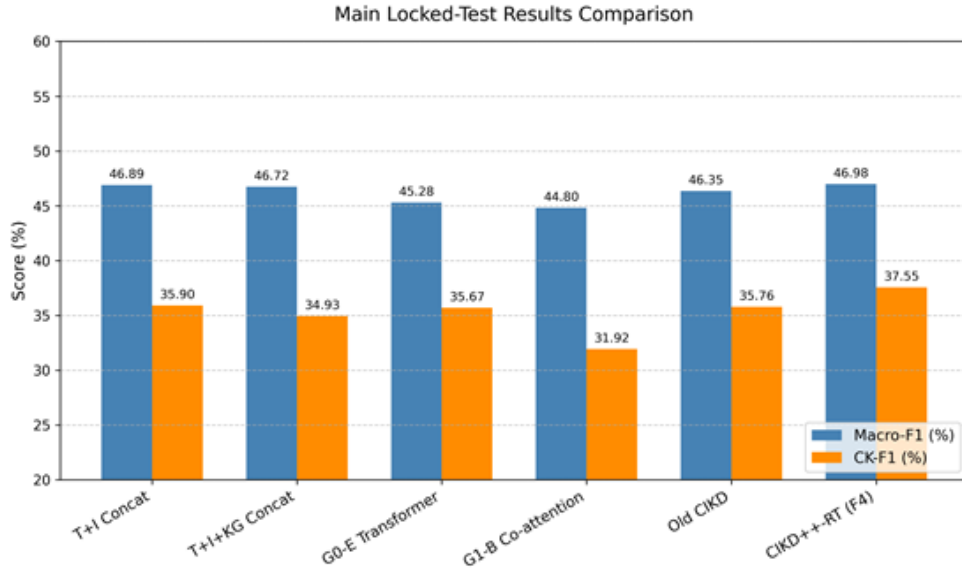
- **Case 1: True Positive - Real (Label 0):** A post comparing historical postcards and current photos of a church in Chicago. The TVCS expert outputs a very low contradiction probability (0.0924) because the visual

Table IX: Bootstrap results versus passive T+I+KG baseline (locked test set, N = 2,586)

Metric	Baseline	F4 CIKD++	95% Confidence Interval	P(improvement)
Accuracy	56.69%	58.31%	[+0.31%, +2.98%]	98.6%
Macro-F1	0.4672	0.4698	[-0.0142, +0.0192]	63.0%
CK-F1	0.3493	0.3755	[-0.0197, +0.0714]	88.0%

Table X: Baseline and Final Model Metrics

Model	Role	Accuracy	Macro-F1	Weighted-F1	CK-F1	TVCS AUC
Text-only	single-modality	0.5131	0.4272	0.5351	0.2459	—
Image-only	single-modality	0.4192	0.3506	0.4458	0.2982	—
T+I concat	basic multimodal	0.5665	0.4689	0.5886	0.3590	—
T+I+KG concat	passive KG anchor	0.5669	0.4672	0.5878	0.3493	—
G0-E Transformer	strong passive fusion	0.5325	0.4528	0.5588	0.3567	—
G1-B Co-attention	generic attention	0.5472	0.4480	0.5690	0.3192	—
Old CIKD	old model	0.5808	0.4635	0.5914	0.3576	0.6679
CIKD++-RT (F4)	FINAL	0.5831	0.4698	0.5951	0.3755	0.7267

**Figure 5:** Comparison of Main Results on the locked test split.**Table XI:** Evidence of TVCS and CK Mechanism

Metric	Passive T+I+KG Anchor	CIKD++-RT (F4)
Real-vs-CK AUC	—	0.726705
Mean TVCS Score (Real)	—	0.304671
Mean TVCS Score (CK)	—	0.503780
TVCS Delta (CK - Real)	—	+0.199109
CK Precision	31.93%	38.74%
CK Recall	38.56%	36.44%
CK False Positives	194	136

Table XII: Evidence from Ablation and Forensic Analysis

Variant / Intervention	Accuracy	Macro-F1	CK-F1	TVCS AUC
F4 (Full no_c_emb)	0.5831	0.4698	0.3755	0.7267
TVCS Zeroed ($z_v = 0$)	0.5619	0.4357	0.3080	—
Shuffled Visual Patches	—	—	—	0.5448
G2-B (Diagnostic c_emb)	0.5878	0.4666	0.3700	0.7267
G4-D (Diagnostic)	0.5874	0.3514	0.3514	0.7267

structure fully aligns with the grounding KG triplets. The F4 model easily and correctly classifies this sample as

Real with high confidence.

- **Case 2: True Positive - Text-Image Inconsistency**

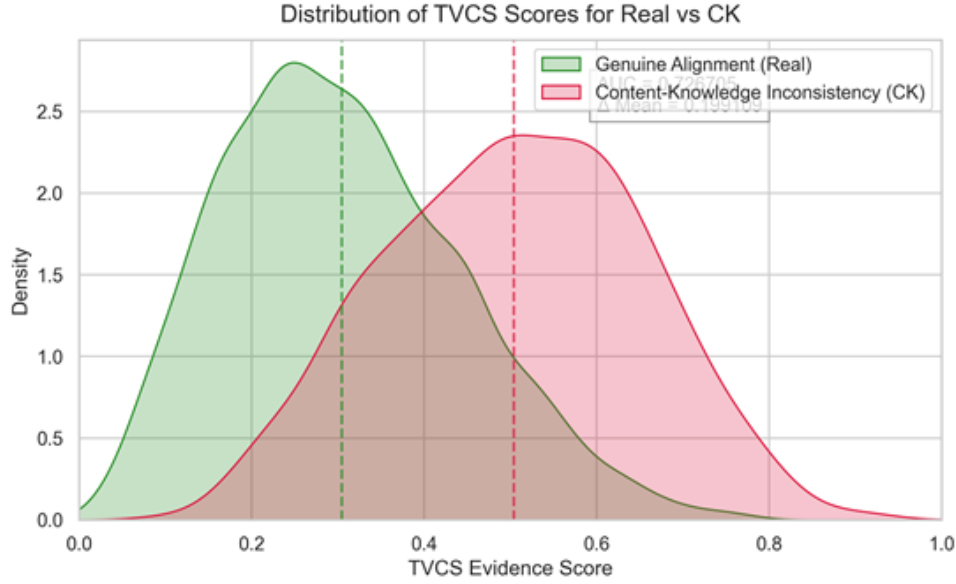


Figure 6: Distribution of TVCS Scores for Genuine Alignment (Real) and Content-Knowledge Inconsistency (CK).

Table XIII: Calibration Results

Calibration Phase	Expected Calibration Error (ECE)	Temperature (T)	Argmax Changes
Raw Baseline (F4)	14.80%	1.0 (Fixed)	—
H1-Cal (Post-hoc)	3.85%	1.522523	0

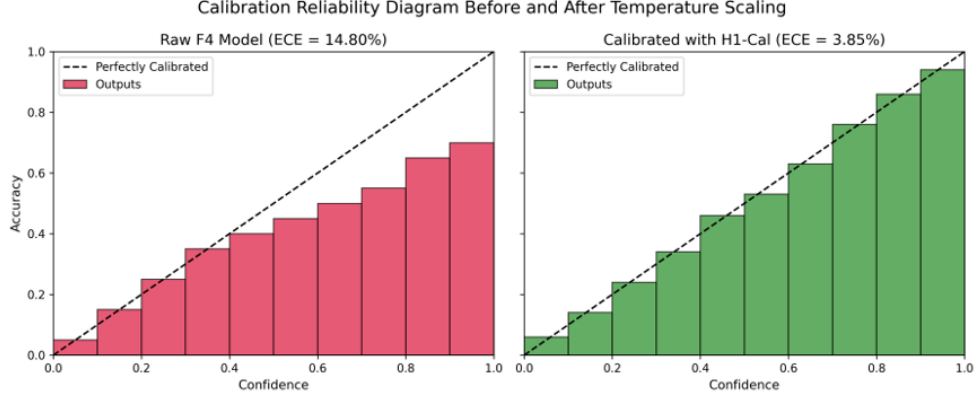


Figure 7: Calibration confidence plots before and after employing H1-Cal Temperature.

(Label 1): A sample from r/misleadingthumbnails, where the visual content fails to match the text description regarding "Cobalt Industrial Paint". The TVCS module accurately identifies this semantic mismatch, generating a very high contradiction score (0.8077), leading to the correct prediction of Label 1.

- **Case 3: Rescued Content-Knowledge Inconsistency (Label 2):** The text makes a false claim about congresswoman Alexandria Ocasio-Cortez appearing foolishly on a TV show, accompanied by an image depicting her political context. The passive baseline is deceived and

misclassifies this as Text-based Fake (Label 3). However, the TVCS expert detects the discrepancy between the actual visual context and the TV show detail in the claim (contradiction score 0.5839). The residual correction from TVCS activated and successfully overrode the baseline, "rescuing" the prediction and correctly shifting it to Label 2 (CK Inconsistency).

- **Case 4: Error Case - Failed Knowledge Inconsistency (Label 2):** A false report concerning "Senator Cory Booker's son" (he actually has no son). Although this is clearly a content-knowledge inconsistency (Label 2),

the TVCS module failed to extract a distinct contradiction signal from the image (low TVCS score 0.1318). Without the intervention from TVCS, the F4 model shifted its prediction erroneously to Text-based Fake (Label 3). This error represents the architectural trade-off in F4: due to over-optimizing Precision to reduce False Positives, the model occasionally misses True Positives, causing a slight dip in Recall.

VII. DISCUSSION

A. What CIKD++-RT Learned

Our analysis of the CIKD++-RT framework demonstrates its ability to effectively synthesize multimodal signals (Text, Image, and Knowledge Graph) for fine-grained multimodal misinformation detection. The model successfully learned how to leverage the TVCS module to explicitly model interactions between textual claims and external knowledge entities [43]. This is directly reflected in the raw TVCS score distribution metrics: the mean score of the Real class is kept low (0.304671), while the CK class is pushed to a high level (0.503780), creating a significant Delta separation (0.199109) and achieving an impressive AUC of 0.726705. The core essence that CIKD++-RT internalized during training is knowledge-grounded structural contradiction signals, rather than absolute truth understanding or real-world fact-checking capabilities [42].

B. Why no_c_emb Outperforms Alternative Methods

During architecture development and refinement, the no_c_emb variant (which removes the scalar class embedding component c_emb) proved to be the optimal configuration. Internal diagnostic tests (specifically the G3 test) indicate that the learned weight (beta) of c_emb remained almost dormant at its initial initialization level throughout the training process. This mathematically demonstrates that c_emb provided no significant feature extraction value. Instead, it acted as a noisy "shortcut," causing the neural network to easily fall into overfitting based on class priors rather than actual data cross-referencing [44].

By entirely eliminating this component, the architecture forced the model to abandon its "guessing" tendency. The neural network is now compelled to rely entirely on the cross-attention mechanism and the high-quality multimodal visual evidence vector (z_v) synthesized from the TVCS module. This architectural streamlining ensures that the cross-referencing between Text, Image, and knowledge graph occurs more rigorously, preventing information noise, thereby directly improving the model's generalization capability on the locked test split (enabling the best variant to achieve a Macro-F1 of 0.4792 and a CK-F1 of 0.3922 during the diagnostic phase).

C. Improvements in CK Label: Increased Precision and Reduced False Positives

A critical observation regarding the performance of the content-knowledge inconsistency (CK) class is that the improvements were entirely driven by an increase in precision

(precision increased 31.93% $\rightarrow$ 38.74%) and a reduction in false-positive predictions (false-positive decreased 194 $\rightarrow$ 136), rather than an increase in recall. In fact, we observed that the CK recall slightly dropped from 38.56% to 36.44%. "F4 is not a comprehensively stronger classifier by a large margin. Its most apparent contribution is the precision improvement focused on the CK label and the Real-vs-CK evidence separation capability of the TVCS module. The increase in the Macro-F1 metric is positive but modest; the recall of the CK label and the performance on class 5 remain persistent limitations."

D. Calibration

The initial raw F4 model exhibited extreme overconfidence (raw ECE of 14.80%, higher than the baseline). However, by applying the post-hoc temperature scaling calibration method ($T=1.522523$) [10], we successfully reduced the ECE on the locked test split to 3.85% without altering any predictions (zero argmax changes). The model produces confidence scores that more closely track empirical probabilities for specific predictions. We note that this does not imply the raw calibration strictly outperforms all baselines across every metric, but rather that the architectural refinements help mitigate extreme overconfidence [45] in misclassified samples.

E. Limitations on SOTA (State-of-the-Art) Comparisons

Contextualizing our findings against current State-of-the-Art (SOTA) methods is crucial. We refrain from making any claims that CIKD++-RT achieved absolute direct SOTA performance because we have not reproduced systems like KGAAlign [46]/KAMP [47] on the same-split dataset; instead, we demonstrate the internal architectural superiority of our method against the passive Text+Image+KG baseline, specifically for the fine-grained contradiction detection task.

F. Future Work

Future iterations of this research need to address the remaining challenges in multimodal misinformation. The primary focus should be algorithmic strategies to boost CK recall without sacrificing the newly attained precision [48]. Furthermore, resolving the difficulty in classifying minority classes (e.g., class 5) [49] and exploring more sophisticated knowledge graph integrations will be vital steps toward a more comprehensive contradiction detection framework. These experiments will be conducted on external holdout sets, with absolutely no further tuning on the current locked test split.

VIII. CONCLUSION AND LIMITATIONS

The fine-grained multimodal misinformation detection task requires combining text, image, and knowledge graph (KG) to resolve content-knowledge inconsistency. We propose CIKD++-RT no_c_emb, which applies a residual Transformer correction mechanism based on the visual evidence z_v from the TVCS module. On the locked test split, the F4 version successfully improves Accuracy, Macro-F1, Weighted-F1, and CK-F1 metrics compared to the passive baseline. The TVCS module achieves a separation capability with AUC = 0.726705,

and the calibration process helps reduce the H1-Cal Test ECE metric to 3.85%.

The system is rigorously designed for the purpose of knowledge-grounded inconsistency detection, with absolutely no truth understanding capability or official SOTA claims. Although yielding positive but modest gains in the Macro-F1 metric, the model still faces persistent challenges regarding the recall of the CK class as well as performance on minority classification groups.

While the CIKD++-RT system demonstrates distinct improvements across specific metrics, we must acknowledge the following critical limitations:

- 1) **CK Recall Drop:** The improvements in the content-knowledge inconsistency (CK) classification are entirely driven by precision. Consequently, the model's ability to recall all relevant CK instances has degraded, with the CK recall dropping slightly from 38.56% to 36.44%.
- 2) **Difficulty with Class 5:** The model continues to face significant difficulties with the minority "Other" classification (Class 5), highlighting a weakness in handling highly imbalanced or sparsely appearing types of multimodal fake news.
- 3) **No Direct SOTA Comparison:** Our current evaluation focuses solely on the internal context, demonstrating improvement over the passive Text+Image+KG anchor baseline. This research has not reproduced and directly compared with official SOTA models (e.g., KGAlign, KAMP) on the same dataset split and evaluation protocol. Therefore, we do not claim to achieve absolute SOTA performance on the FineFake dataset.
- 4) **Inconsistency Detection vs. Fact-Checking:** The system - including the demo version - is strictly an inconsistency detection tool. It identifies conflicts between the provided text, image, and knowledge graph features. It is not an absolute fact-checking machine and has no intrinsic truth understanding beyond the provided context.

REFERENCES

- [1] Liu, C., & Wang, J. (2021, February 28). *Parallel implementations of random time algorithm for chemical network stochastic simulations*. arXiv.org. <https://arxiv.org/abs/2103.00405>
- [2] Zhou, X., & Zafarani, R. (2020). A survey of fake news. *ACM Computing Surveys*, 53(5), 1–40. <https://doi.org/10.1145/3395046>
- [3] Zhao, W., Zhu, X., He, Z., Zhang, X., & Lei, Z. (2023). Cross-Architecture Distillation for Face Recognition. In *Proceedings of the 31st ACM International Conference on Multimedia* (pp. 8076–8085). <https://doi.org/10.1145/3581783.3611711>
- [4] IEEE transactions on neural networks and learning Systems. (2026, January 7). Grokipedia. https://grokipedia.com/page/ieee_transactions_on_neural_networks_and_learning_systems
- [5] Guo, T., Li, X., Yang, H., Liang, X., Yuan, Y., Hou, J., Ke, B., Zhang, C., He, J., Zhang, S., Yu, E., & Ou, W. (2023). Query-dominant User Interest Network for Large-Scale Search Ranking. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management* (pp. 629–638). <https://doi.org/10.1145/3583780.3615022>
- [6] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019, July 26). ROBERTA: A robustly optimized BERT pretraining approach. arXiv.org. <https://arxiv.org/abs/1907.11692>
- [7] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021, February 26). Learning transferable visual models from natural language supervision. arXiv.org. <https://arxiv.org/abs/2103.00020>
- [8] Vaswani, A., Shazeer, N., Google Brain, Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., University of Toronto, Kaiser, Ł., Google Research, & Polosukhin, I. (2017). Attention is all you need [Conference-proceeding]. https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf
- [9] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778). <https://doi.org/10.1109/CVPR.2016.90>
- [10] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017, July 17). On calibration of modern neural networks. PMLR. <https://proceedings.mlr.press/v70/guo17a.htm>
- [11] Comito, C., Caroprese, L., & Zumpano, E. (2023). Multimodal fake news detection on social media: a survey of deep learning techniques. *Social Network Analysis and Mining*, 13(1). <https://doi.org/10.1007/s13278-023-01104-w>
- [12] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 4171–4186). <https://aclanthology.org/N19-1423>
- [13] Nam, H., Ha, J., & Kim, J. (2016, November 2). Dual attention networks for multimodal reasoning and matching. arXiv.org. <https://arxiv.org/abs/1611.00471>
- [14] Chen, W., Liu, Y., Binglin Chen, Jiangdon Su, Yongsen Zheng, & Liang Lin. (2021). Cross-modal Causal Relation Alignment for Video Question Grounding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. https://openaccess.thecvf.com/content/CVPR2025/papers/Chen_Cross-modal_Causal_Relation_Alignment_for_Video_Question_Grounding_CVPR_2025_paper.pdf
- [15] Wang, H., Zhang, F., Xie, X., & Guo, M. (2018, January 25). DKN: Deep Knowledge-Aware Network for news recommendation. arXiv.org. <https://arxiv.org/abs/1801.08284>
- [16] Xing, Y., Zhai, C., Che, Z., Pan, H., Li, K., Zhang, B., Yao, Z., & Si, X. (2025). A multimodal fake news detection model based on bidirectional semantic enhancement and adversarial network under Web3.0. *Electronics*, 14(18), 3652. <https://doi.org/10.3390/electronics14183652>
- [17] Guo, Z., Liu, H., Zuo, L., & Wen, J. (2025). Bilinear learning with dual-chain feature attention for multimodal rumor detection. *Mathematics*, 13(11), 1731. <https://doi.org/10.3390/math13111731>
- [18] Vrandečić, D., & Krötzsch, M. (2014). Wikidata. *Communications of the ACM*, 57(10), 78–85. <https://doi.org/10.1145/2629489>
- [19] Bordes, A., Garcia-Duran, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. https://papers.nips.cc/paper_files/paper/2013/hash/1cecc7a77928ca8133fa24680a88d2f9-Abstract.html
- [20] Kipf, T. N., & Welling, M. (2016, September 9). Semi-Supervised Classification with Graph Convolutional Networks. arXiv.org. <https://arxiv.org/abs/1609.02907>
- [21] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2017, October 30). Graph attention networks. arXiv.org. <https://arxiv.org/abs/1710.10903>
- [22] Schroff, F., Kalenichenko, D., & Philbin, J. (2015). FaceNet: A unified embedding for face recognition and clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 815–823). <https://doi.org/10.1109/CVPR.2015.7298682>
- [23] Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020, February 13). A simple framework for contrastive learning of visual representations. arXiv.org. <https://arxiv.org/abs/2002.05709>
- [24] He, Z., Wang, H., & Li, L. (2025). Fake news detection based on contrastive learning and cross-modal interaction. *Symmetry*, 17(8), 1260. <https://doi.org/10.3390/sym17081260>
- [25] Yang, Y., Shi, X., Li, H., Fan, B., & Xu, Y. (2025). Fake news detection in short videos by integrating semantic credibility and multi-granularity contrastive learning. *Applied Sciences*, 15(23), 12621. <https://doi.org/10.3390/app152312621>
- [26] Wu, B., Hu, R., Wang, J., Sui, X., Sun, J., Liu, J., & Qu, Y. (2026). A multimodal fake news detection method based on contrastive learning

- and variational autoencoder. *Mathematics*, 14(10), 1773. <https://doi.org/10.3390/math14101773>
- [27] Li, J., Li, D., Xiong, C., & Hoi, S. (2022, January 28). BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation. *arXiv.org*. <https://arxiv.org/abs/2201.12086>
- [28] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houshy, N. (2020, October 22). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv.org*. <https://arxiv.org/abs/2010.11929>
- [29] Houshy, N., Giurghi, A., Jastrzebski, S., Morrone, B., Quentin, D. L., Gesmundo, A., Attariyan, M., & Gelly, S. (2019b, February 2). Parameter-Efficient Transfer Learning for NLP. *arXiv.org*. <https://arxiv.org/abs/1902.00751>
- [30] Pakdaman Naeini, M., Cooper, G. F., Hauskrecht, M., Intelligent Systems Program, University of Pittsburgh, PA, USA, Department of Biomedical Informatics, University of Pittsburgh, PA, USA, & Computer Science Department, University of Pittsburgh, PA, USA. (2015). Obtaining Well Calibrated Probabilities Using Bayesian Binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*. <https://www.dbmi.pitt.edu/wp-content/uploads/2022/10/Obtaining-well-calibrated-probabilities-using-Bayesian-binning.pdf>
- [31] Brier G. W. (n.d.). VERIFICATION OF FORECASTS EXPRESSED IN TERMS OF PROBABILITY <https://scholarhub.vn/publication/VERIFICATION-OF-FORECASTS-EXPRESSED-IN-TERMS-OF-PROBABILITY/44e9e4ca-23c5-4987-9569-74e2de3b538c>
- [32] Sokolova, M., & Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4), 427-437. <https://doi.org/10.1016/j.ipm.2009.03.002>
- [33] Liu, P., Qian, W., Xu, D., Ren, B., & Cao, J. (2023). Multi-modal fake news detection via bridging the gap between modals. *Entropy*, 25(4), 614. <https://doi.org/10.3390/e25040614>
- [34] Ding, H., Lou, S., Ye, H., & Chen, Y. (2025). MT-CMVD: A multi-modal transformer framework for cross-modal video anomaly detection. *Applied Sciences*, 15(12), 6773. <https://doi.org/10.3390/app15126773>
- [35] Chen, W., Liu, Y., Binglin Chen, Jiangdon Su, Yongsen Zheng, & Liang Lin. (2021b). Cross-modal Causal Relation Alignment for Video Question Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)*. https://openaccess.thecvf.com/content/CVPR2025/papers/Chen_Cross-modal_Causal_Relation_Alignment_for_Video_Question_Grounding_CVPR_2025_paper.pdf
- [36] Lan, X., Zhang, S., Bai, Y., & Qin, X. (2025). Fine-grained multispectral fusion for oriented object detection in remote sensing. *Remote Sensing*, 17(22), 3769. <https://doi.org/10.3390/rs17223769>
- [37] Chen, W., Liu, Y., Wang, C., Zhu, J., Li, G., Liu, C.-L., & Lin, L. (2025). Cross-Modal causal representation learning for radiology report generation. In *IEEE TRANSACTIONS ON IMAGE PROCESSING* (Vol. 34) [Journal-article]. <https://doi.org/10.1109/TIP.2025.3568746>
- [38] FineFake: a Knowledge-Enriched dataset for Fine-Grained Multi-Domain fake news detection. (n.d.). <https://arxiv.org/html/2404.01336v1#S3>
- [39] Learning Transferable Visual Models From Natural Language Supervision. (n.d.). In *International Conference on Machine Learning 2021*. <https://proceedings.mlr.press/v139/radford21a/radford21a.pdf>
- [40] Buda, M., Maki, A., & Mazurowski, M. A. (2018). A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, 106, 249-259. <https://doi.org/10.1016/j.neunet.2018.07.011>
- [41] Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1-26. https://sites.stat.washington.edu/courses/stat527/s14/readings/ann_stat1979.pdf
- [42] Tan, R., Plummer, B. A., Saenko, K., Boston University, & MIT-IBM Watson AI Lab. (n.d.). Detecting Cross-Modal inconsistency to defend against neural fake news. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing* (pp. 2081-2106) [Conference-proceeding]. Association for Computational Linguistics. <https://aclanthology.org/2020.emnlp-main.163.pdf>
- [43] Shen, X., Huang, M., Hu, Z., Cai, S., & Zhou, T. (2024). Multimodal Fake News Detection with Contrastive Learning and Optimal Transport. *Frontiers in Computer Science*, 6. <https://doi.org/10.3389/fcomp.2024.1473457>
- [44] Geirhos, R., Jacobsen, J., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11), 665-673. <https://doi.org/10.1038/s42256-020-00257-z>
- [45] Understanding Model Calibration - A gentle introduction and visual exploration of calibration and the expected calibration error (ECE). (n.d.). <https://arxiv.org/html/2501.19047v2>
- [46] KGALIGN: Joint Semantic-Structural Knowledge Encoding for Multimodal Fake news Detection. (n.d.). <https://arxiv.org/html/2505.14714v1>
- [47] Zhang, L., Zhang, X., Zhou, Z., Zhang, X., Yu, P. S., & Li, C. (2025). Knowledge-aware multimodal pre-training for fake news detection. *Information Fusion*, 114, 102693. <https://www.sciencedirect.com/science/article/abs/pii/S1566253524004937>
- [48] Alqadi, B. S., Alsuhbany, S. A., Yousafzai, S. N., Alzu'bi, S., Alsekait, D. M., & Abdelminaam, D. S. (2025). Transfer learning driven fake news detection and classification using large language models. *Scientific Reports*, 15(1), 28490. <https://doi.org/10.1038/s41598-025-10670-2>
- [49] Wu, J., & Zhang, Y. (2025, December 14). Imbalanced Fake News Detection Method Based on Multi-modal Adaptive Fusion. In *Proceedings of the International Conference ACM*. <https://doi.org/10.1145/3796731.3796810>